[S07] Retrieval-Augmented Generation (RAG): Grounding LLM Responses in Documents

TOPIC AREA Generative AI / LLM

I. Learning Objectives and Guiding Question

Explain how a retriever is combined with a generator, measure retrieval performance, and assess whether generated responses are supported by evidence.

Guiding question: Does providing external documents eliminate hallucinations in language models?

II. Required Tasks

1. Explain parametric and non-parametric memory, distinguish RAG from fine-tuning and from simply placing all source documents into one prompt.
2. Draw a pipeline: ingestion → chunking → embedding/keyword index → retrieval → prompt augmentation → answer generation. Discuss potential failure points at each stage.
3. Build a small knowledge corpus comprising roughly 5–10 pages of public or properly licensed materials. Prepare 15–20 questions with reference answers and verifiable sources.
4. Experiment with a lightweight retrieval system using TF-IDF/BM25 or cosine similarity between embeddings. Evaluate top-k hit rate or Recall@k using at least two chunk sizes or two values of k.
5. If an LLM is available, compare answers with and without retrieval for the same questions. Otherwise, focus on retrieval analysis and demonstrate how retrieved evidence is inserted into a prompt.
6. Categorize failures: missing retrieval, incorrect retrieved documents, irrelevant citations, and generation beyond the evidence. Propose verification and document-update mechanisms.

III. Equations / Representations to Explain

RAG-Sequence: $p(y|x) \approx \sum_\{z \in top-k\}\, p\eta(z|x) \cdot p\theta(y|x,z)$

Student need to define every symbol, specify appropriate units/value ranges and modeling assumptions, and provide a complete step-by-step worked example.

IV. Minimum Expected Outputs

RAG pipeline; at least one worked question → retrieved chunks → answer/citation example; a Recall@k table; and at least three analyzed failures.

V. 5. Required and Recommended Research Papers

Required: Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020.

https://papers.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html

Reading focus: Read the distinction between parametric and non-parametric memory, and compare RAG-Sequence with RAG-Token.

Recommended: Schneider, J. (2024). Explainable Generative AI (GenXAI): a survey, conceptualization, and research agenda. Artificial Intelligence Review 57, 289. https://doi.org/10.1007/s10462-024-10916-x

Reading focus: Read the discussion of evidence attribution and verification in generative AI.

VI. Format and Submission Requirements

- Team size: three students are recommended. Suggested preparation time: approximately two weeks after the topic is assigned.
- Submit a 6–10-page research-style report in Vietnamese on A4 paper (excluding references and appendices), organized as: Abstract, Introduction, Background/Paper Review, Method/Worked Example, Results and Discussion, Conclusion, and References.
- Prepare 8–12 content slides for a 15-minute seminar presentation followed by approximately 5 minutes of Q&A. Every team member must present and answer at least one question.
- Read all papers marked “Required” in this assignment. Explain the material in your own words and distinguish findings reported in the papers from examples or experiments developed by your team.

- Programming-based experiments are encouraged but not generally mandatory. A worked numerical example, an illustrative model, and an evidence-supported comparison are mandatory.
- Cite all figures, tables, datasets, equations, and AI tools used. Independently verify all claims suggested by AI-generated content.

VII. Seminar Assessment Rubric (100 points)

Assessment criterion	Requirements for full marks	Points
Problem statement and objectives	Establish relevant context, a clear research question, and an appropriate scope.	5
Reading and synthesis of research papers	Accurately explain the original papers' objectives, contributions, methods, and findings, with appropriate source comparison.	15
Explanation of algorithms and equations	Define all variables and assumptions, describe algorithmic steps, and provide at least one correct numerical example or diagram.	15
Evidence, comparison, or experimentation	Include an appropriate baseline or reference, verifiable data or examples, and clearly explained figures and tables.	10
Critical analysis and limitations	Discuss strengths, weaknesses, and conditions of applicability. Avoid overstating conclusions based on a small demonstration.	10
Slide quality and citations	Prepare 8–12 clear, legible slides, cite sources for figures/equations and relevant papers, and avoid verbatim copying.	10
Seminar presentation	Deliver a coherent presentation within 15 minutes, explain examples clearly, and ensure balanced participation.	15
Q&A and critical	Answer questions accurately, justify technical decisions,	20

discussion	and ensure that each member answers at least one question.	

Performance levels for each criterion: Excellent (100%), Good (75%), Satisfactory (50%), and Unsatisfactory (0–25%) of the available points, depending on the quality of evidence submitted.